



SGGS Institute of Engineering and Technology, Vishnupuri, Nanded
ISE - I (Semester-I) 2025-26

B. Tech.
Computer Vision

Computer Science & Engineering
PEC-CS416

Date: 18/August/2025

Time: 11.00 - 11.45

Max. Marks : 20

- Note: i. Attempt all questions
ii. Assume suitable data if necessary
iii. Use of non-programmable calculator is permitted.
iv. This paper contains 2 pages.

Q.N.		Question	Marks	CO	BT
Q.1	1	Which of the following best describes computer vision? A. A branch of AI that enables machines to understand and interpret visual information from the world B. A field focused on designing computer hardware for image processing C. A technique for generating random images D. A subset of robotics only	1	CS416.1	BT1
	2	Which of the following is NOT a typical task of computer vision? A. Image classification B. Object detection C. Speech recognition D. Semantic segmentation	1	CS416.1	BT2
	3	What is the main purpose of histogram equalization? A. To reduce image resolution B. To enhance image contrast C. To compress the image size D. To reduce image noise	1	CS416.1	BT1
	4	In histogram equalization, the transformation function is based on: A. The mean of pixel intensities B. The cumulative distribution function (CDF) of intensities C. Fourier transform of the image D. Image gradient	1	CS416.1	BT2
	5	Histogram equalization may cause: A. Contrast loss in already high-contrast images B. Increase in image noise visibility C. Over-enhancement in some regions D. All of the above	1	CS416.1	BT3
	6	Connected component labeling is mainly used to: A. Compress an image B. Identify and label distinct objects in a binary image C. Increase image resolution D. Perform histogram equalization	1	CS416.2	BT2
	7	Which of the following are commonly used types of pixel connectivity? A. 2-connectivity and 3-connectivity B. 4-connectivity and 8-connectivity C. 5-connectivity and 10-connectivity D. Row and column connectivity	1	CS416.2	BT2
	8	The main purpose of the Fourier Transform in image processing is: A. To enhance image contrast B. To represent an image in the frequency domain C. To compress images D. To increase resolution	1	CS416.1	BT2
	9	The high-frequency components in the Fourier spectrum correspond to: A. Image smoothness B. Fine details and sharp edges C. Overall brightness D. Low contrast regions	1	CS416.1	BT3

	10	A low-pass filter is used to: A. Sharpen the image B. Smooth the image by reducing high-frequency noise C. Detect edges D. Enhance fine details	1	CS416.1	BT3
Q.2		What is a pixel? Define spatial resolution.	2	CS416.1	BT1
Q.3		Define connected components in a binary image.	2	CS416.2	BT2
Q.4		What does the magnitude of the Fourier Transform represent? What does the phase of the Fourier Transform represent?	2	CS416.1	BT2
Q.5		Define 2D DFT and IDFT along with its equations.	2	CS416.1	BT2
Q.6		Which filter is best suited for removing salt-and-pepper noise? Explain its process.	2	CS416.1	BT3

Course Outcomes (CO)

CS416.1 - Describe different image representations, their mathematical representation and data structures used.

CS416.2 - Classify different Boundary Detection algorithms.

CS416.3 - Create a 3D object from a given set of images.

CS416.4 - Detect a moving object in video using the concept of motion analysis.

CS416.5 - Recognize the object using the concept of computer vision.

Bloom's Taxonomy (BT)

BT1- Remember, BT2- Understand, BT3- Apply, BT4- Analyze, BT5- Evaluate, BT6- Create



**Shri Guru Gobind Singhji Institute of Engineering
and Technology, Nanded-431606**

ISE-I Exam

Subject: **Distributed Computing**

Date: 19/08/2025

Time: 11.00 To 11.45

Code: **PCC-CS404**

Max.Marks: 20

Instructions:

1. All questions are compulsory.
2. Assume suitable data if necessary.

Q.1 Write in brief to solve the Following (Any Two)

10

- a. Why is it not always a good idea to aim at implementing the highest degree of transparency possible?
- b. What is DCE RPC ? Explain the steps to design a client and server application using DCE RPC.
- c. Define Distributed System and explain design issues Of DOS.
- d. Define transparency and different types of transparencies in the distributed system.

Q.2 a. What is scalability? Discuss the scalability techniques in distributed systems.

05

b. Explain the difference between the primitives mpi_bsend and mpi_isend in MPI.

05

End



**SGGS Institute of Engineering and Technology,
Vishnupuri, Nanded**

In Semester Evaluation- I (Semester-I) 2025

Final Year B. Tech.

Computer Science & Engineering

TCP/IP Networking

PCC-CS402

Time: 11:00 to 11:45

Max. Marks : 20

Date:

21/08/20245

- Note: i. Attempt all questions
ii. Assume suitable data if necessary
iii. Use of non-programmable calculator is permitted.
iv. Figures to the right indicate Marks

- Q.1 Change the following IP addresses from Binary notation to dotted-decimal-notation. 04
- a. 11000010 10001011 01001011 00001111
b. 11000011 10001100 00101011 01010101
- Q.2 Find the Netid and Hostid for the following IP addresses: 06
- a. 81.60.99.10
b. 160.80.91.67
c. 210.16.4.111
- Q.3 An organization is granted the class B block 176.40.0.0. The administrator wants to create 128 subnets. Find the subnet mask, the number of addresses in each subnet and the range of addresses in the first two and last two subnets. 06
- Q.4 Find the First and Last address of the block for the following Classless IP addresses. 04
- a. 167.199.170.82/27
b. 140.120.84.24/20